vitamins should be supplied regularly to our body. The missing numbers in B vitamins are once considered as vitamins but no longer considered as such, and the numbers that were assigned to them now form the gaps.

**Table 14.2: Vitamins, their Sources, Functions and their Deficiency disease**

| Vitamin | Sources | Functions | Deficiency Disease |
|---|---|---|---|
| Vitamin A (Retinol) | Liver oil, Fish, Carrot, Milk, spinach and fruits such as Papaya and mango | Vision and growth | Night blindness, Xerophthalmia Keratinisation of skin |
| Vitamin $B_1$ (Thiamine) | Yeast, Milk, Cereals, Green vegetables, Liver, Pork | Co – enzyme in the form of Thiamine pyro phosphate (TPP) in glycolysis | Beri – Beri (peripheral nerve damage) |
| Vitamin $B_2$ (Riboflavin) | Soybean, Green vegetable Yeast, Egg white, Milk, Liver kidney | Co enzyme in the form of FMN (Flavin mono nucleotide)and FAD (Flavin adenine dinucleotide) in redox reactions | Cheilosis (lesions of corner of mouth, lips and tongue) |
| Vitamin $B_3$ (Niacin) | Cereals, Green leafy vegetables, Liver, Kidney | Co enzyme in the form of NAD and $NADP^+$ in redox reactions. | Pellagra (photo sensitive dermatitis) |
| Vitamin $B_5$ (Pantothenic acid) | Mushroom, Avocado, Egg yolk, Sunflower oil | Part of coenzyme A in carbohydrate protein and Fat metabolism | Inadequate growth |
| Vitamin $B_6$ (Pyridoxine) | Meat, Cereals, Milk, Whole grains, Egg. | Co enzyme in amino acid metabolism, formation of Heme in Hemoglobin | Convulsions |
| Vitamin $B_7$ (Biotin) | Liver, kidney, Milk, Egg yolk, Vegetables, Grains | Co enzyme in fatty acid Biosynthesis | Depression, Hair loss, muscle pain. |
| Vitamin $B_9$ (Folic acid) | Egg, Meat, Beet root, Leafy vegetables, Cereals, Yeast | Nucleic acid synthesis, maturation of red blood cells | Megaloblastic anaemia |



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